Tropical Gradient Descent

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Abstract

We propose a gradient descent method for solving optimization problems arising in settings of tropical geometry—a variant of algebraic geometry that has become increasingly studied in applications such as computational biology, economics, and computer science. Our approach takes advantage of the polyhedral and combinatorial structures arising in tropical geometry to propose a versatile approach for approximating local minima in tropical statistical optimization problems—a rapidly growing body of work in recent years. Theoretical results establish global solvability for 1-sample problems and a convergence rate matching classical gradient descent. Numerical experiments demonstrate the method's superior performance over classical descent for tropical optimization problems which exhibit tropical convexity but not classical convexity. We also demonstrate the seamless integration of tropical descent into advanced optimization methods, such as Adam, offering improved overall performance.

Keywords: tropical quasi-convexity; tropical projective torus; gradient descent; phylogenetics.

1 Introduction

Where algebraic geometry studies geometric properties of solution sets of systems of multivariate polynomials, *tropical geometry* restricts to the case of polynomials defined by linearizing operations, where the "sum" of two elements is their maximum and the "product" of two elements is their sum. Evaluating polynomials with these operations results in piecewise linear functions. Tropical geometry is a relatively young field of pure mathematics established in the 1990s (with roots dating back further) and has recently become an area of active interest in computational and applied mathematics for its relevance to the statistical analyses of phylogenetic trees (Monod *et al.*, 2018), the problem of dynamic programming in computer science (Maclagan & Sturmfels, 2015), and in mechanism design for two-player games in game theory (Lin & Tran, 2019). Following the identification of the tropical Grassmannian and the space of phylogenetic trees (Speyer & Sturmfels, 2004) in particular, there has been surge of research (Barnhill *et al.*, 2024b; Lin & Yoshida, 2018; Comăneci & Joswig, 2024; Monod *et al.*, 2018) which necessitates the development of new computational techniques for optimisation in tropical geometric settings.

The training of machine learning models and the vast majority of computational tasks in statistics entail solving optimization problems; this paper is motivated by statistical optimization problems in tropical geometric settings. Initial research into tropical statistics considered optimization problems such as the identification of Fermat–Weber points which, due to the piecewise linear nature of tropical geometry, can be re-framed as a linear program (Comăneci & Joswig, 2024; Lin & Yoshida, 2018). Unfortunately, the complexity of these linear programs often scales with the size of our dataset, so gradient descent is more commonly implemented in practice (Barnhill *et al.*, 2024a,b). More recently—following the identification of ReLU neural networks and tropical rational functions (Zhang *et al.*, 2018)—more sophisticated neural network optimization problems are being studied in the tropical setting (Pasque *et al.*, 2024; Yoshida *et al.*, 2024), which necessitates the use of gradient methods for optimization in the tropical setting. In this paper, we address the natural question—given the need for gradient methods in tropical data science, how can we tailor classical gradient methods to the tropical setting? We provide a foundational tropical gradient descent framework to solve the optimization problems which arise in the computation of a wide range of tropical statistics, an area of active interest in applied tropical geometry.

The remainder of this paper is organized as follows. The following section contains the necessary foundations of tropical geometry, an introduction to tropical location problems (Comăneci, 2024), and the array of statistical optimization tasks studied in this paper. Section 3 contains the theoretical contributions of this paper; we formulate steepest descent with respect to the tropical norm and present the theoretical guarantees for tropical descent, most notably its convergence for a wide class of tropically quasi-convex functions and a convergence rate of $O(1/\sqrt{k})$. In Section 4, we perform a comprehensive numerical study of our proposed tropical descent method and its performance in the statistical optimization problems of interest. We conclude with a discussion of possible future work on tropical gradient methods in Section 5.

2 Background and Preliminaries

In this section, we overview the background on tropical geometry and present the statistical optimization problems which we study in the tropical setting.

2.1 Tropical Geometry

Here, we review the algebraic and geometric structure of the tropical projective torus, the state space for phylogenetic data and the domain of our tropical statistical optimization problems.

Definition 1 (Tropical Algebra)**.** The *tropical algebra* is the semiring $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty\}$ with the addition and multiplication operators—tropical addition and tropical multiplication, respectively—given by

$$a \boxplus b = \max\{a, b\}, \quad a \odot b = a + b.$$

The additive identity is $-\infty$ and the multiplicative identity is 0. Tropical subtraction is not defined; tropical division is given by classical subtraction.

The tropical algebra defined above is sometimes referred to as the *max-plus algebra*; we can define the min-plus algebra analogously, taking $\min\{a, b\} =: a \oplus b$ as the additive operation on $\mathbb{R} \cup \{\infty\}$. These are algebraically equivalent through negation. Unless specified, we use the max-plus convention as this is better suited for phylogenetic statistics on the tropical projective torus (Speyer & Sturmfels, 2004).

Definition 2 (Tropical Projective Torus)**.** The n-dimensional *tropical projective torus* is a quotient space constructed by endowing $\mathbb{R}^N$ with the equivalence relation

$$\mathbf{x} \sim \mathbf{y} \Leftrightarrow \exists\, a : \ \mathbf{x} = a \odot \mathbf{y}; \tag{1}$$

it is denoted by $\mathbb{R}^N/\mathbb{R}\mathbf{1}$. The generalized Hilbert projective metric, also referred to as the *tropical metric*, is given by

$$d_{\mathrm{tr}}(\mathbf{x}, \mathbf{y}) = \max_i\{x_i - y_i\} - \min_i\{x_i - y_i\} = \max_{i,j}\{x_i - y_i - x_j + y_j\}.$$

This metric is induced by the tropical norm, which is given by

$$\|\mathbf{x}\|_{\mathrm{tr}} = \max_i x_i - \min_i x_i.$$

Remark 3. While a single point in the tropical projective torus is given by some ray $\mathbf{x} + \mathbb{R}\mathbf{1}$, we note that there is a unique representative of this equivalence class whose coordinates sum to zero. In taking such representatives, we can identify the tropical projective torus $\mathbb{R}^N/\mathbb{R}\mathbf{1}$ and the hyperplane $\mathcal{H} = \{\mathbf{x} \in \mathbb{R}^N : \sum x_i = 0\}$. Throughout this paper, we use this homeomorphism to visualize the tropical projective torus (e.g., Figures 1 and 2).

The tropical projective torus is the ambient space containing the tropical Grassmannian, which is equivalent to the space of phylogenetic trees (Speyer & Sturmfels, 2004). This has sparked the study of tropical data science, formalizing statistical techniques which respect the tropical geometry of the state space (Monod *et al.*, 2018; Yoshida, 2022).

The generalized Hilbert projective metric defined above is widely accepted and preferred for the development of geometric statistical tools on the tropical projective torus (Yoshida, 2022). However, recent work has shown theoretical advantages to the use of an asymmetric metric (Comăneci & Joswig, 2024).

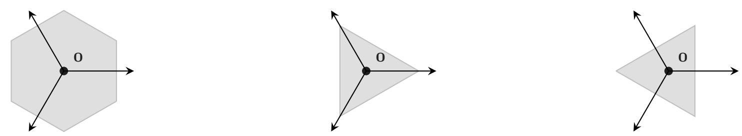

Figure 1: The d_{tr}, $d_{\triangle_{\min}}$ and $d_{\triangle_{\max}}$ tropical unit balls in $\mathbb{R}^3/\mathbb{R}\mathbf{1} \cong \mathcal{H} = \{\sum x_i = 0\}$.

Definition 4 (Tropical Asymmetric Distance)**.** The *tropical asymmetric distances* on $\mathbb{R}^N/\mathbb{R}\mathbf{1}$ are given by

$$d_{\triangle_{\min}}(\mathbf{a}, \mathbf{b}) := \sum_i (b_i - a_i) - N \min_j (b_j - a_j),$$
$$d_{\triangle_{\max}}(\mathbf{a}, \mathbf{b}) := N \max_j (b_j - a_j) - \sum_i (b_i - a_i).$$

Remark 5. The asymmetric distances act as an ℓ_1 norm on the tropical projective torus, while the symmetric metric d_{tr} acts as an ℓ_∞ norm. We note that $N d_{\text{tr}} = d_{\triangle_{\min}} + d_{\triangle_{\max}}$. Figure 1 shows the balls for each tropical metric of interest.

We now define the primary geometric objects of interest in the tropical projective torus; hyperplanes, lines, and convex hulls. We use the max-convention for these definitions, though we note each has a min-convention equivalent.

Definition 6 (Tropical Hyperplane)**.** The *tropical hyperplane* $\mathcal{H}_{\mathbf{a}}$ defined by the linear form $a_1 \odot x_1 \boxplus \cdots \boxplus a_N \odot x_N$ is given by

$$\mathcal{H}_{\mathbf{a}} = \{\mathbf{x} : \exists\, i \neq j \text{ s.t. } a_i + x_i = a_j + x_j = \max\{a_k + x_k\}\}.$$

Definition 7 (Tropical Line Segment)**.** For any two points $\mathbf{a}, \mathbf{b} \in \mathbb{R}^N/\mathbb{R}\mathbf{1}$, the *tropical line segment* between $\mathbf{a}$ and $\mathbf{b}$ is the set

$$\gamma_{\mathbf{ab}} = \{\alpha \odot \mathbf{a} \boxplus \beta \odot \mathbf{b} \mid \alpha, \beta \in \mathbb{R}\}$$

with tropical addition taken coordinate-wise.

Tropical line segments define a unique geodesic path between any two points, but general geodesics are not uniquely defined on the tropical projective torus.

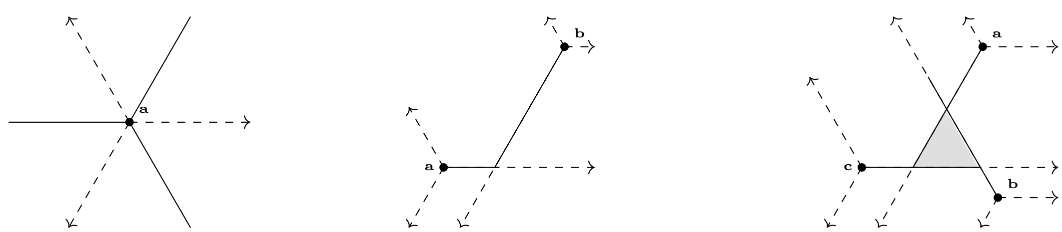

Figure 2: A tropical hyperplane, tropical line segment and tropical convex hull in $\mathbb{R}^3/\mathbb{R}\mathbf{1} \cong \mathcal{H} = \{\mathbf{x} : \sum x_i = 0\}$. The dashed lines show coordinate directions.

Definition 8 (Tropical Convex Hull (Develin & Sturmfels, 2004))**.** A set $S \subset \mathbb{R}^N/\mathbb{R}\mathbf{1}$ is max-*tropically convex* if it contains $\alpha \odot \mathbf{x} \boxplus \beta \odot \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in S$ and $\alpha, \beta \in \mathbb{R}$. For a finite subset $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_K\} \subset \mathbb{R}^N/\mathbb{R}\mathbf{1}$, the *max-plus tropical convex hull* of X is the set of all max-tropical linear combinations of points in X,

$$\operatorname{tconv}_{\max}(X) := \{\alpha_1 \odot \mathbf{x}_1 \boxplus \cdots \boxplus \alpha_K \odot \mathbf{x}_K \mid \alpha_1, \ldots, \alpha_K \in \mathbb{R}\}.$$

Similarly, the *min-tropical convex hull* of X, $\operatorname{tconv}_{\min}(X)$, is the set of all min-tropical linear combinations of points in X.

A study of tropically convex functions requires an understanding of both max-tropical and min-tropical convexity, as we will see in our following consideration of tropical location problems.

2.2 Tropical Location Problems

Tropical location problems are a new avenue of research, motivated by the optimization of statistical loss functions; they involve the minimization over some dataset of some loss function which heuristically increases with distance from some kernel. Here we outline the relevant definitions of tropical location problems and their theoretical behavior as presented by Comăneci (2024). We introduce the problem of tropical linear regression (Akian *et al.*, 2023) as motivation.

2.2.1 Motivating Example

The problem of tropical linear regression looks to find the best-fit tropical hyperplane which minimizes the maximal distance to a set of data points. While this is an inherently statistical optimization problem, it has recently been proven to be polynomial-time equivalent to solving mean payoff games (Akian *et al.*, 2023).

A tropical hyperplane is uniquely defined by a cone point $\mathbf{t} \in \mathbb{R}^N/\mathbb{R}\mathbf{1}$, allowing us to formulate the best-fit tropical hyperplane as an optimization problem over $\mathbb{R}^N/\mathbb{R}\mathbf{1}$.

Definition 9 (Tropical Linear Regression (Akian *et al.*, 2023))**.** Let $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_K\} \subset \mathbb{R}^N/\mathbb{R}\mathbf{1}$ be our dataset. The *tropical linear regression problem* finds the vertex $\mathbf{t} \in \mathbb{R}^N/\mathbb{R}\mathbf{1}$ of the best-fit hyperplane:

$$\min_{\mathbf{t}\in\mathbb{R}^N/\mathbb{R}\mathbf{1}} f(\mathbf{t}) = \max_{k\leq K} d_{\mathrm{tr}}(\mathbf{x}, \mathcal{H}_{\mathbf{t}}) = \max_{k\leq K} (\mathbf{x}_k - \mathbf{t})_i - (\mathbf{x}_k - \mathbf{t})_j$$

where i, j are such that $\forall\, \ell \neq i, j : (\mathbf{x}_k - \mathbf{t})_i \geq (\mathbf{x}_k - \mathbf{t})_j \geq (\mathbf{x}_k - \mathbf{t})_\ell$.

In Figure 3, we see a heat map of this objective function over the tropical projective plane for a sample of size 3. We see immediately that the objective is not convex. The linear regression objective has sparse gradient almost everywhere, which produces valleys. These properties are considered unfavourable extreme cases for gradient methods, but are particularly common in tropical optimization problems. While the tropical linear regression problem is not convex, it is quasi-convex along tropical line segments; that is, its sub-level sets are tropically convex.

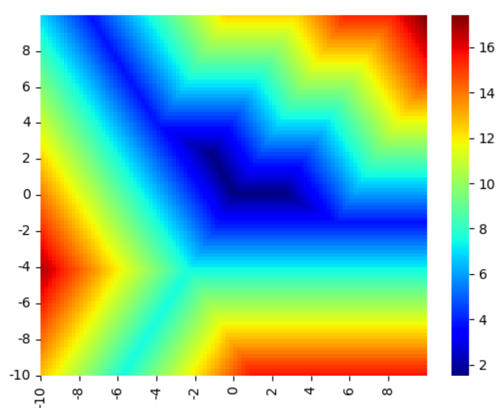

Figure 3: A heat map of a 3-sample linear regression objective on $\mathbb{R}^3/\mathbb{R}\mathbf{1} \cong \mathcal{H} = \{\mathbf{x} : \sum x_i = 0\}$.

2.2.2 Quasi-Convexity of Tropical Functions

In formalizing tropical location problems, the work of Comăneci (2024) first considers functions which increase along tropical geodesics from a kernel $\mathbf{v}$; that is, for any point $\mathbf{t}$ on a geodesic from $\mathbf{v}$ to $\mathbf{w}$, we have that $f(\mathbf{t}) \leq f(\mathbf{w})$. This property is equivalent to star-convexity of sublevel sets. To formally state this, we define (oriented) geodesic segments and $\triangle_{\min}$-star convexity of sets.

Definition 10 (Oriented Geodesic Segment)**.** The *(oriented) geodesic segment* between $\mathbf{a}, \mathbf{b} \in \mathbb{R}^N/\mathbb{R}\mathbf{1}$, is given by $[\mathbf{a}, \mathbf{b}]_{\triangle_{\min}} \coloneqq \{\mathbf{x} \in \mathbb{R}^N/\mathbb{R}\mathbf{1} : d_{\triangle_{\min}}(\mathbf{a}, \mathbf{x}) + d_{\triangle_{\min}}(\mathbf{x}, \mathbf{b}) = d_{\triangle_{\min}}(\mathbf{a}, \mathbf{b})\}$.

The oriented geodesic segment is the union of all geodesics from $\mathbf{a}$ to $\mathbf{b}$ with respect to $d_{\triangle_{\min}}$.

Definition 11 ($\triangle_{\min}$-Star Convex Set)**.** A set $K \subseteq \mathbb{R}^N/\mathbb{R}\mathbf{1}$ is a $\triangle_{\min}$*-star-convex* set with kernel $\mathbf{v}$ if, for every $\mathbf{w} \in K$, we have $[\mathbf{v}, \mathbf{w}]_{\triangle_{\min}} \subseteq K$.

Using the above definitions, we can characterize a loss function which is increasing along tropical geodesics as a function with $\triangle_{\min}$-star-convex sublevel sets. The following definition gives a closed form expression for such functions, while Theorem 13 proves their equivalence.

Definition 12 ($\triangle_{\min}$-Star-Quasi-Convex Functions (Comăneci, 2024))**.** A function $f : \mathbb{R}^N/\mathbb{R}\mathbf{1} \to \mathbb{R}$ is $\triangle_{\min}$*-star-quasi-convex* with kernel $\mathbf{v}$ if $f(\mathbf{x}) = \hat{\gamma}(\mathbf{x} - \mathbf{v})$ for some $\gamma : \mathbb{R}^N_{\geq 0} \to \mathbb{R}$ which is increasing in every coordinate, where we define $\hat{\gamma}(\mathbf{x}) \coloneqq \gamma(\mathbf{x} - (\min_i x_i)\mathbf{1})$.

Theorem 13 (Theorem 17 of (Comăneci, 2024))**.** *A continuous function $f : \mathbb{R}^N/\mathbb{R}\mathbf{1} \to \mathbb{R}$ is $\triangle_{\min}$-star-quasi-convex with kernel $\mathbf{v}$ if and only if all of its non-empty sub-level sets are $\triangle_{\min}$-star convex with kernel $\mathbf{v}$.*

Remark 14. We can define $\triangle_{\max}$-star-quasi-convex functions similarly, but we also note that a function $f(\mathbf{t})$ is a $\triangle_{\max}$-star-quasi-convex function if and only if $f(-\mathbf{t})$ is a $\triangle_{\min}$-star-quasi-convex function.

These $\triangle_{\min}$-star-convex functions act as a loss function with respect to a single data point. For example, the tropical metric d_{tr} is both $\triangle_{\min}$ and $\triangle_{\max}$-star-quasi-convex. The distance to a hyperplane $d_{\mathrm{tr}}(\mathbf{x}, \mathcal{H}_{\mathbf{t}})$ is $\triangle_{\max}$-star-quasi-convex in $\mathbf{x}$, but $\triangle_{\min}$-star-quasi-convex in $\mathbf{t}$.

As a final remark, we note that $\triangle_{\min}$-star-quasi-convex functions are a special case of tropically quasi-convex functions by the following lemma. We limit our theoretical considerations to $\triangle_{\min}$-star-quasi-convex functions to utilize their closed form expression given in Definition 12, however numerical experiments demonstrate strong performance of our methodology for more general tropical quasi-convex functions.

Proposition 15 (Proposition 8 of (Comăneci, 2024))**.** *Any $\triangle_{\min}$-star-convex set is* min*-tropically convex. Hence, any $\triangle_{\min}$-star-quasi-convex function is* tropically quasi-convex *in that it has tropically convex sublevel sets.*

2.2.3 Tropical Location Problems

The $\triangle_{\min}$-star-convex functions defined above are generally a measure of closeness to the kernel $\mathbf{v}$, but in minimizing a statistical loss function we look at closeness to a sample of points. A *location problem* (Laporte *et al.*, 2019) measures the closeness to a sample rather than a single kernel.

Definition 16 (Min(/max)-Tropical Location Problem (Comăneci, 2024))**.** Let $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_K\} \subset \mathbb{R}^N/\mathbb{R}\mathbf{1}$ be our dataset. For $k \leq K$, let h_k be a $\triangle_{\min}$-star-quasi-convex function with kernel $\mathbf{x}_k$, and let $g : \mathbb{R}^K \to \mathbb{R}$ be increasing in every coordinate. A *min-tropical location problem* is the minimization of an objective function $f : \mathbb{R}^N/\mathbb{R}\mathbf{1} \to \mathbb{R}$ given by $f = g(h_1, \ldots, h_K)$. A *max-tropical location problem* is defined similarly, where the h_k are $\triangle_{\max}$-star-quasi-convex.

The main result of (Comăneci, 2024) is the following: tropical location problems contain a minimum in the tropical convex hull of its dataset X.

Theorem 17 (Theorem 20 of (Comăneci, 2024))**.** *Let f be the objective function of a min-tropical location problem. Then there is a minimum of f belonging to* $\mathrm{tconv}_{\max}(X)$*. Conversely, max-tropical location problems have minima belonging to* $\mathrm{tconv}_{\min}(X)$*.*

We can now revisit our motivating example (Definition 9), showing that tropical linear regression is not just a min-tropical location problem, but also demonstrates tropical quasi-convexity in that it has tropically convex sublevel sets.

Proposition 18. *The tropical linear regression problem is a* min*-tropical location problem with tropically convex sublevel sets.*

Proof

$$\begin{aligned} f(\mathbf{t}) &= \max_{k\leq K} \min_j [\max_i [x_{ki} - t_i] - x_{kj} + t_j] \\ &= \max_{k\leq K} \min_j [t_j - x_{kj} - \min_i [t_i - x_{ki}]] \end{aligned}$$

This is a min-tropical location problem with:

$$\begin{aligned} g(\mathbf{h}) &= \max_{k\leq K} h_k, \\ h_k(\mathbf{t}) &= \hat{\gamma}(\mathbf{t} - \mathbf{x}_k), \\ \gamma(\mathbf{t}) &= \min_j t_i. \end{aligned}$$

This g and γ are increasing in each coordinate and each h_k has kernel $\mathbf{x}_k$, satisfying the conditions for a min-tropical location problem.
As the h_k are $\triangle_{\min}$-star-convex, their sublevel sets are min-tropically convex by Proposition 15. This is preserved when taking a maximum over samples, so f also has min-tropically convex sublevel sets. □

As mentioned in Proposition 15, $\triangle_{\min}$-star-quasi-convexity implies tropical quasi-convexity. However, the tropical linear regression problem is a case in which the reverse implication does not hold; in general, the linear regression loss function is not $\triangle_{\min}$-star-quasi-convex.

2.3 Tropical Data Science

While young, the field of tropical data science is motivated by the crucial question of statistical learning from the increasing volume of phylogenetic tree data (Kahn, 2011). In contrast to the BHV (Billera *et al.*, 2001) or Robinson–Foulds metrics (Robinson & Foulds, 1981), the tropical interpretation of tree space provides both interpretable geometry and computational efficiency (Steel & Penny, 1993; Monod *et al.*, 2018). Many of the statistical methods defined for the tropical setting are framed as optimisation problems (Barnhill *et al.*, 2024b), and in this subsection we review some such tropical statistical optimization problems which we use to test our tropical descent methodology. We note the varying degrees of convexity demonstrated by each of them. While we restrict our considerations through the rest of this paper to statistics which arise from tropical location problems, in Appendix A we discuss other tropical statistical optimization problems.

2.3.1 Centrality Statistics

Our first problems of interest, Fermat–Weber points and Fréchet means, are centrality statistics for data on a general metric space; Fermat–Weber points (Drezner *et al.*, 2004) act as a generalized median, while Fréchet means (Fréchet, 1948) are a generalization of the mean.

Definition 19 (Tropical Fermat–Weber Points (Lin & Yoshida, 2018))**.** Let $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_K\} \subset \mathbb{R}^N/\mathbb{R}\mathbf{1}$ be our dataset. A solution to the following optimization problem is a *tropical Fermat–Weber point*:

$$\min_{\mathbf{t}\in\mathbb{R}^N/\mathbb{R}\mathbf{1}} f(\mathbf{t}) = \frac{1}{K}\sum_{k\leq K} d_{\mathrm{tr}}(\mathbf{x}_k, \mathbf{t}).$$

Definition 20 (Tropical Fréchet means (Monod *et al.*, 2018))**.** Let $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_K\} \subset \mathbb{R}^N/\mathbb{R}\mathbf{1}$ be our dataset. A solution to the following optimization problem is a *tropical Fréchet mean*:

$$\min_{\mathbf{t}\in\mathbb{R}^N/\mathbb{R}\mathbf{1}} f(\mathbf{t}) = \left[\frac{1}{K}\sum_{k\leq K} d_{\mathrm{tr}}(\mathbf{x}_k, \mathbf{t})^2\right]^{1/2}.$$

The Fermat–Weber and Fréchet mean problems are both max-tropical and min-tropical location problems, as the tropical metric d_{tr} is both $\triangle_{\min}$ and $\triangle_{\max}$-star-quasi-convex. In fact, they are also both classically convex and hence act as a benchmark against which we can test our gradient methods.

2.3.2 Wasserstein Projections

The next statistical optimization problem was motivated by Cai & Lim (2022), and looks to identify a tropical projection which minimizes the Wasserstein distance between samples in different tropical projective tori.

Definition 21 (Tropical Wasserstein Projections (Talbut *et al.*, 2023))**.** Let $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_K\} \subset \mathbb{R}^N/\mathbb{R}\mathbf{1}$ be a dataset in one space, and $Y = \{\mathbf{y}_1, \ldots, \mathbf{y}_K\} \subset \mathbb{R}^M/\mathbb{R}\mathbf{1}$ be a dataset in a different space. Let $J_1, \ldots, J_M$ be a non-empty partition of $[N]$. Then the *tropical p-Wasserstein projection problem* is the minimization of the following objective function:

$$\min_{\mathbf{t}\in\mathbb{R}^N/\mathbb{R}\mathbf{1}} f_p(\mathbf{t}) = \left(\frac{1}{K}\sum_{k\leq K} \|[\max_{i\in J_j}(\mathbf{x}_k - \mathbf{t})_i]_j - \mathbf{y}_k\|_{\mathrm{tr}}^p\right)^{1/p}$$

The *tropical ∞-Wasserstein projection problem* is the minimization of:

$$\min_{\mathbf{t}\in\mathbb{R}^N/\mathbb{R}\mathbf{1}} f_\infty(\mathbf{t}) = \max_{k\leq K} \|[\max_{i\in J_j}(\mathbf{x}_k - \mathbf{t})_i]_j - \mathbf{y}_k\|_{\mathrm{tr}}$$

In contrast to the work by Lee *et al.* (2022) which looks to solve the optimal transport problem for intrinsic measures on $\mathbb{R}^N/\mathbb{R}\mathbf{1}$, this p-Wasserstein projection problem is computing an optimal projection for empirical samples.

Proposition 22. *The tropical p, ∞-Wasserstein projections are min-tropical location problems, and the ∞-Wasserstein projection has* min*-tropically convex sublevel sets.*

Proof Let $\mathbf{z}_k = (x_{ki} - y_{kj})_{i\leq N}$ where j is such that J_j is the unique partition set containing i. We then have that

$$\begin{aligned}
\|[\max_{i\in J_j}(\mathbf{x}_k - \mathbf{t})_i]_j - \mathbf{y}_j\|_{\mathrm{tr}} &= \max_b \max_{a\in J_b}[z_{ka} - t_a] - \min_j \max_{i\in J_j}[z_{ki} - t_i] \\
&= \max_a[z_{ka} - t_a] + \max_j \min_{i\in J_j}[t_i - z_{ki}] \\
&= \max_j[\min_{i\in J_j}[t_i - z_{ki} - \min_a[t_a - z_{ka}]]]
\end{aligned}$$

The p, ∞-Wasserstein projections can then be written as

$$\begin{aligned}
f_p(\mathbf{t}) &= \left(\frac{1}{K}\sum_{k\leq K}\left(\max_j[\min_{i\in J_j}[t_i - z_{ki} - \min_a[t_a - z_{ka}]]]\right)^p\right)^{1/p} \\
f_\infty(\mathbf{t}) &= \max_{k\leq K}\max_j[\min_{i\in J_j}[t_i - z_{ki} - \min_a[t_a - z_{ka}]]]
\end{aligned}$$

These are max-tropical location problems with:

$$\begin{aligned}
g_p(\mathbf{h}) &= \left(\frac{1}{K}\sum_{k\leq K} h_k^p\right)^{1/p}, \\
g_\infty(\mathbf{h}) &= \max_{k\leq K} h_k, \\
h_k(\mathbf{t}) &= \hat{\gamma}(\mathbf{t} - \mathbf{z}_k), \\
\gamma(\mathbf{t}) &= \max_j \min_{i\in J_j} t_i.
\end{aligned}$$

The g_p, g_∞ and γ are increasing in each coordinate and each h_k has kernel $\mathbf{z}_k$, satisfying the conditions for a min-tropical location problem.
As the h_k are $\triangle_{\min}$-star-convex, their sublevel sets are min-tropically convex by Proposition 15. This is preserved when taking a maximum over samples, so f_∞ also has min-tropically convex sublevel sets. $\square$

Figure 4 shows a heat map for the ∞-Wasserstein projection objectives. As in the tropical linear regression problem, it is piecewise linear, non-convex and has sparse gradient.

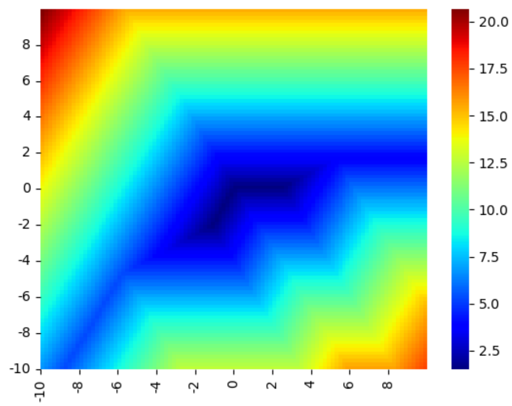

Figure 4: A heat map of a 3-sample ∞-Wasserstein projection objective on $\mathbb{R}^3/\mathbb{R}\mathbf{1} \cong \mathcal{H} = \{\mathbf{x} : \sum x_i = 0\}$.

3 Steepest Descent with the Tropical Norm

This section comprises the theoretical contributions of this paper; we outline our proposed method of tropical descent, and the theoretical convergence guarantees for tropical descent when applied to tropical location problems.

We treat optimization over $\mathbb{R}^N/\mathbb{R}\mathbf{1}$ as optimization over $\mathbb{R}^N$ with the knowledge that f is constant along $\mathbf{1}$-rays. We assume we can compute some gradient-like function $\nabla f : \mathbb{R}^N \to \mathbb{R}^N$ which is satisfying Assumption 23.

Assumption 23. *Let $f = g(h_1, \ldots, h_K)$ be a min-tropical location problem. Then we have that*

$$\nabla f(\mathbf{t})_j < 0 \Rightarrow \exists\, k \text{ s.t. } j \in \mathit{argmin}_i(t_i - x_{ki}).$$

This is a very general assumption, which holds for some of the most natural gradient-like functions. For example, if $\nabla f(\mathbf{t})$ is given by some Clarke gradient, that is, some element of the *generalized Clarke derivative* of f, then Lemma 25 shows that Assumption 23 holds.

Definition 24 (Generalized Clarke Derivative (2.13, I.I of Kiwiel (1985)))**.** We define the set $M_f(\mathbf{t})$ by

$$M_f(\mathbf{t}) = \{\mathbf{z} \in \mathbb{R}^N : \nabla f(\mathbf{y}_l) \to \mathbf{z},\ \text{where } \mathbf{y}_l \to \mathbf{t}, f \text{ differentiable at each } \mathbf{y}_l\}.$$

Then the *generalized Clarke derivative* is given by

$$\partial f(\mathbf{t}) = \operatorname{conv} M_f(\mathbf{t}).$$

Lemma 25. *Suppose $f = g(h_1, \ldots, h_K)$ is min-tropical location problem, and for all $\mathbf{t}$, $\nabla f(\mathbf{t})$ is some element of $\partial f(\mathbf{t})$. Then Assumption 23 holds.*

Proof Suppose that $\nabla f(\mathbf{t})_j < 0$. Then there is a sequence of differentiable $\mathbf{y}_\ell$ satisfying $\nabla f(\mathbf{y}_\ell)_j < 0$, $\mathbf{y}_\ell \to \mathbf{t}$. Hence by the chain rule, as g is increasing in every h_k, there is some h_k which is strictly decreasing in the j^{th} coordinate at $\mathbf{y}_\ell$. WLOG, the same h_k is strictly decreasing in the j^{th} coordinate for all $\mathbf{y}_\ell$. As γ is increasing in every coordinate, $h_k = \hat{\gamma}(\mathbf{y}_\ell - \mathbf{x}_k)$ is strictly decreasing in the j^{th} coordinate if and only if $j \in \operatorname{argmin}_i(y_{\ell i} - x_{ki})$. This is preserved through the limit as $\ell \to \infty$, so we conclude that $j \in \operatorname{argmin}_i[t_i - x_i]$. □

In practice, rather than computing elements of the generalized Clarke derivative, we use PyTorch automatic differentiation (Paszke *et al.*, 2019). The derivatives computed by PyTorch are dependent on the computational representation of a function, but are designed to respect the chain rule. We must therefore ensure that our computational representation of g has non-negative derivatives in each h_k, while each h_k

is computed such that its t_j derivative is negative only if $i \in \operatorname{argmin}_i(t_i - x_{ki})$. For such a computational representation of f, Assumption 23 will hold by the chain rule. For all the tropical location problems we consider in this paper, we find their most natural computational representation gives rise to derivatives ∇f satisfying Assumption 23.

3.1 Proposed Method

In defining tropical descent, we refer back to the motivation behind classical gradient descent—steepest descent directions.

Definition 26 (Steepest Descent Direction (Boyd & Vandenberghe, 2004))**.** A normalized *steepest descent direction* at $\mathbf{t}$ with respect to the norm $\|\cdot\|$ is:

$$\mathbf{d} = \operatorname{argmin}\{\nabla f(\mathbf{t})^\top \mathbf{v} : \|\mathbf{v}\| = 1\}.$$

With respect to the Euclidean norm, the normalized derivative is a steepest descent direction. When we use a the tropical norm, the steepest descent directions are characterized by the following lemma.

Lemma 27. *Suppose $\nabla f \neq \mathbf{0}$. Then the direction $\mathbf{v}$ is a normalized tropical steepest descent direction iff*

$$\begin{aligned} v_i = 1 &\Leftrightarrow \nabla f_i < 0 \\ v_i = 0 &\Leftrightarrow \nabla f_i > 0 \end{aligned}$$

Proof WLOG, we assume $0 \leq v_i \leq 1$. Then

$$\nabla f^\top \mathbf{v} \geq \sum_{i:\, \nabla f_i < 0} \nabla f_i$$

This is achieved if and only if $v_i = 1$ for all i such that $\nabla f_i < 0$, and $v_i = 0$ for all i such that $\nabla f_i > 0$. We note that any such v_i satisfies $\|\mathbf{v}\|_{\mathrm{tr}} = 1$ as $\sum_i \nabla f_i = 0$ by the equivalence relation of the tropical projective torus, and $\nabla f_i \neq \mathbf{0}$ so ∇f has positive and negative coordinates. □

Lemma 27 uniquely defines a descent direction $\mathbf{d}$ if and only if, for every coordinate i, $\nabla f_i \neq 0$. To uniquely define a descent direction for any $\mathbf{t}$, we take the direction $\mathbf{d}$ given by

$$d_i = \begin{cases} 1 & \text{if } \nabla f_i < 0, \\ 0 & \text{otherwise.} \end{cases}$$

We refer to this as a *min-tropical descent* direction. In contrast,

$$d_i = \begin{cases} -1 & \text{if } \nabla f_i > 0, \\ 0 & \text{otherwise.} \end{cases}$$

defines a *max-tropical descent* direction.

Throughout the rest of this paper, we use the min-tropical descent direction as convention, and will specify when the max-tropical descent direction is used instead. Tropical descent therefore refers to the method outlined by Algorithm 1, for some specified step size sequence $(a_n)_{n\geq 1}$, while we refer to classical gradient descent as classical descent.

Algorithm 1 Tropical Descent

for $n <$ max_steps **do**
  $\mathbf{g}_{ni} \leftarrow \nabla f_i(\mathbf{t}_{n-1})$;
  $\mathbf{d}_{ni} \leftarrow \mathbf{1}_{\mathbf{g}_{ni} < 0}$;
  $\mathbf{t}_n \leftarrow \mathbf{t}_{n-1} + a_n \mathbf{d}_n$;

We note that over a large number of steps, Algorithm 1 can produce large t_{ni} values as each coordinate is non-decreasing in n. If this results in overflow or precision errors, a normalisation operation can be included in each step, such as subtracting $\frac{1}{N}\sum_i t_{ni}$ from each t_{ni}. The function f is invariant under such a translation due to the equivalence relation on the tropical projective torus.

In (Barnhill *et al.*, 2024a,b), classical descent is implemented for the computation of Fermat–Weber points; this is the most natural choice as the Fermat–Weber problem is classically convex. However, the linear regression and Wasserstein projection problems lack classical convexity, and hence do not consistently produce accurate solutions via classical descent. It is these tropically convex problems for which tropical descent is necessary.

3.2 Convergence Guarantees

In this section we prove the theoretical guarantees of tropical descent for min-tropical location problems. We show that tropical descent must converge to the tropical convex hull of the data, and hence will necessarily find a minimum of a $\triangle_{\min}$-star-convex problem. Throughout this section, we assume the step sizes a_n satisfy $a_n \to 0, \sum a_n = \infty$.

We first highlight the key difference in stability between classical descent and tropical descent; if some component of the derivative is locally zero, tropical descent cannot converge in that neighbourhood.

Proposition 28. *Let $\mathbf{t}_n$ be some sequence defined by tropical descent. Assume that for all n, $\nabla f(\mathbf{t}_n) \neq \mathbf{0}$. Suppose in some bounded open U, ∇f_i is identically 0. Then no point in $\operatorname{int} U$ is a convergence point for $\mathbf{t}_n$; in particular, if $\mathbf{t}_m \in U$ there is some $n > m$ such that $\mathbf{t}_n \notin U$.*

Proof Assume otherwise. Then for all $n \geq m$, we have $\nabla f(\mathbf{t}_n)_i = 0$, so t_{ni} is constant. At each step, we increase at least one coordinate by a_n, so as $n \to \infty$:

$$\sum_j t_{nj} \geq \sum_j t_{mj} + \sum_{n \geq m} a_n \to \infty.$$

Therefore $\|\mathbf{t}_n\|_{\text{tr}}$ will become arbitrarily large, as $\max_j t_{nj} \to \infty$ while $\min_j t_{nj} \leq t_{mi}$. Then we must have left U; contradiction. □

As a result of the proposition above, local minima such as the valleys in Figures 3 and 4 can be stable with respect to classical descent but unstable with respect to tropical descent.
We next prove the main convergence result of this work; tropical descent must converge to the tropical convex hull of the data points.

Theorem 29. *Suppose f is a min-tropical location problem with respect to the dataset $X = \{\mathbf{x}_1, \dots, \mathbf{x}_K\}$. Let $(\mathbf{t}_n)_{n\geq 1}$ be a sequence of points defined by tropical descent, such that for all n: $\nabla f(\mathbf{t}_n) \neq 0$. Let $V = \operatorname{tconv}_{\max}(\mathbf{x}_1, \dots, \mathbf{x}_K)$, and define*

$$\Delta_j(\mathbf{t}) = \sum_{k\in[K]} \sum_{i\in[N]} [t_j - x_{kj} - t_i + x_{ki}]^+.$$

For any $\epsilon > 0$, let M_1 be such that for all $m \geq M_1$, $a_m < \epsilon/N$. Let M_2 be such that for all $m \geq M_2$ $s_m \geq \max_i \Delta_i(\mathbf{t}_{M_1}) + s_{M_1} - K\epsilon$, where s_m is the partial sum of the step size sequence a_m. Then for all $m \geq M_2$ we have:

$$d_{\text{tr}}(\mathbf{t}_n, V) \leq \epsilon.$$

Proof For each $i \leq N$, we define the functions

$$\delta_j(\mathbf{t}) = \min_{k\in[K]} \left[\sum_{i\in[N]} [t_j - x_{kj} - t_i + x_{ki}]^+ \right].$$

Claim 30. *For all $\mathbf{t} \in \mathbb{R}^N/\mathbb{R}\mathbf{1}$, we have $d_{\text{tr}}(\mathbf{t}, V) \leq \max_j \delta_j(\mathbf{t})$.*

Proof As discussed by Maclagan & Sturmfels (2015), there is a well-defined metric projection map π_V onto V given by:

$$\pi_V(\mathbf{t}) = \lambda_1 \odot \mathbf{x_1} \boxplus \cdots \boxplus \lambda_K \odot \mathbf{x}_K,$$
$$\text{where } \lambda_k = \max\{\lambda \in \mathbb{R} : \lambda \odot \mathbf{x}_k \boxplus \mathbf{t} = \mathbf{t}\} = \min_i(t_i - x_{ki}).$$

The distance from $\mathbf{t}$ to V is then given by $d(\mathbf{t}, \pi_V(\mathbf{t}))$, and we have:

$$\begin{aligned} d(\mathbf{t}, V) &= d(\mathbf{t}, \pi_V(t)) \\ &= \|(\max_k(\min_i(t_i - x_{ki}) - t_j + x_{kj})_{j \le N}\|. \end{aligned}$$

We note that for all j, k, the term $\min_i(t_i - x_{ki}) - t_j + x_{kj}$ is negative. Hence:

$$\begin{aligned} d(\mathbf{t}, V) &\le 0 - \min_j(\max_k(\min_i(t_i - x_{ki}) - t_j + x_{kj})) \\ &\le \max_j(\min_k(t_j - x_{kj} - \min_i(t_i - x_{ki}))) \\ &\le \max_j \delta_j(\mathbf{t}). \end{aligned}$$

The claim is proved. □

We will now prove that for all $m \ge M_2$, we have that $\delta_j(\mathbf{t}_m) \le \epsilon$.

Claim 31. *If $\delta_i(\mathbf{t}_m) = 0$ and $m \ge M_1$, then $\delta_i(\mathbf{t}_{m+1}) < \epsilon$.*

Proof For all $k \in [K], j \in [N]$, we have that $t_{mj} - x_{kj} - t_{mi} + x_{ki}$ is 1-Lipschitz in $\mathbf{t}$ (with respect to d_{tr}). Therefore δ_j is N-Lipschitz in $\mathbf{t}$. By the tropical descent rule:

$$\begin{aligned} \delta_j(t_{m+1}) &\le \delta_j(\mathbf{t}_m) + N d_{\text{tr}}(\mathbf{t}_m, \mathbf{t}_{m+1}) \\ &= 0 + N a_m \\ &< \epsilon. \end{aligned}$$

The claim is proved. □

Claim 32. *If $\delta_j(\mathbf{t}_m) > 0$ then $\delta_j(\mathbf{t}_{m+1}) \le \delta_j(\mathbf{t}_m)$.*

Proof Suppose $\delta_j(\mathbf{t}_m) > 0$. Then $\forall k$, $t_{mj} - x_{kj} > \min_i(t_{mi} - x_{ki})$, so by Lemma 25 we have $\nabla f_j \ge 0$. The tropical steepest descent direction therefore has $d_{mj} = 0$. We conclude that

$$\begin{aligned} \forall\, k, i : &\quad t_{(m+1)j} - x_{kj} - t_{(m+1)i} + x_{ki} \le t_{mj} - x_{kj} - t_{mi} + x_{ki}, \\ \Rightarrow &\quad \delta_j(\mathbf{t}_{m+1}) \le \delta_j(\mathbf{t}_m). \end{aligned}$$

The claim is proved. □

Claim 33. *For some $M_1 \le m \le M_2$, $\delta_j(\mathbf{t}_m) < \epsilon$.*

Proof We note that $K\delta_j(\mathbf{t}) \le \Delta_j(\mathbf{t})$; hence if $\Delta_j(\mathbf{t}_{M_1}) < K\epsilon$ then we are done.

We assume $\delta_j(\mathbf{t}_m) > \epsilon$ for all $M_1 \le m \le M_2$, and prove the result by contradiction. Then as in the proof of the previous claim, at each step we have $\nabla f_j \ge 0$, $d_{mj} = 0$ and t_{mj} is fixed for $M_1 \le m \le M_2$.

We have assumed that for each m, $\nabla f \neq \mathbf{0}$ and so there is some i such that $\nabla f_i < 0$, and by Lemma 25, there is some k such that $i \in \operatorname{argmin}_\ell t_{m\ell} - v_{k\ell}$. Hence:

$$\begin{aligned} \forall\, \ell : &\quad t_{mj} - v_{kj} - t_{mi} + v_{ki} \ge t_{mj} - v_{kj} - t_{m\ell} + v_{k\ell}, \\ \Rightarrow &\quad t_{mj} - v_{kj} - t_{mi} + v_{ki} \ge \delta_j(\mathbf{t}_m)/N, \\ &\quad > \epsilon/N, \\ &\quad \ge a_m. \end{aligned}$$

Hence this term in the summation of $\Delta_j(\mathbf{t}_m)$ decreases by a_m, while other terms are non-increasing as t_{mj} is fixed.

Noting that $K\delta_j(\mathbf{t}) \leq \Delta_j(\mathbf{t})$, we conclude

$$\begin{aligned} K\delta_j(\mathbf{t}_{M_2}) &\leq \Delta_j(\mathbf{t}_{M_2}), \\ &\leq \Delta_j(\mathbf{t}_{M_1}) - s_{M_2} + s_{M_1}, \\ &\leq K\epsilon. \end{aligned}$$

Contradiction. So the claim is proved. □

Finally, from the claims above, we conclude that $\forall m \geq M_2, i \in [N]$, we have $\delta_i(\mathbf{t}_m) < \epsilon$, and so by our first claim, we have $d_{\mathrm{tr}}(\mathbf{t}_m, V) \leq \epsilon$. □

As proven by Comăneci (2024), tropical location problems have minima in the tropical convex hull of the data. The theorem above says that according to tropical descent, only such minima can be stable. We cannot do much better in terms of convergence results for general location problems; they can have disconnected sublevel sets so we would not expect gradient methods to find a global minimum. We would require some degree of convexity to guarantee global solvability, which is exactly what the following corollary gives us; tropical descent will necessarily converge to a minimum of any $\triangle_{\min}$-star-quasi-convex function.

Corollary 34. *Let $f(\mathbf{t})$ be a $\triangle_{\min}$-star-quasi-convex function, and let $(\mathbf{t}_n)_{n\geq 1}$ be a sequence of points defined by tropical descent such that for all n: $\nabla f(\mathbf{t}_n) \neq 0$. Then there is some global minimum $\mathbf{t}^*$ such that for all $m \leq M_2$, $d_{\mathrm{tr}}(\mathbf{t}_m, \mathbf{t}^*) \leq \epsilon$ where M_2 is as in Theorem 29.*

As a further corollary, we note that we can achieve the same result for max-tropical location problems by using max-tropical descent directions.

Corollary 35. *Suppose f is a max-tropical location problem with respect to the dataset $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_K\}$. Let $(\mathbf{t}_n)_{n\geq 1}$ be a sequence of points defined via max-tropical descent directions, such that for all n: $\nabla f(\mathbf{t}_n) \neq 0$. Let $V = \mathrm{tconv}_{\min}(\mathbf{x}_1, \ldots, \mathbf{x}_K)$, and define*

$$\Delta_i'(\mathbf{t}) = \sum_{k\in[K]} \sum_{j\in[N]} [t_j - x_{kj} - t_i + x_{ki}]^+.$$

For any initial $\mathbf{t}_0$ and $\epsilon > 0$, let M_1 be such that for all $m \geq M_1$, $a_m \leq \epsilon/N$. Let M_2 be such that for all $m \geq M_2$ $s_m \geq \max_i \Delta_j'(\mathbf{t}_{M_1}) + s_{M_1} - K\epsilon$. Then for all $m \geq M_2$ we have:

$$d_{\mathrm{tr}}(\mathbf{t}_n, V) \leq \epsilon.$$

Remark 36. The results above indicate convergence to V or $\mathbf{t}^*$ in the tropical topology of $\mathbb{R}^N/\mathbb{R}\mathbf{1}$, but $\mathbf{t}_n$ cannot not converge in the Euclidean topology of $\mathbb{R}^N$ due to the divergent partial sums of α_n.

We conclude by establishing the convergence rate of tropical descent in practice; we take step sizes given by $a_m = \alpha m^{-1/2}\|\nabla f(\mathbf{t}_m)\|_{\mathrm{tr}}$, and assume derivatives have bounded tropical norm as is the case for piecewise linear 1-Lipschitz objective functions h_k. In this case, we prove convergence at a rate of $O(\sqrt{m})$.

Proposition 37. *Suppose f is a min-tropical location problem as in Theorem 29. Assume that there is some $L \in \mathbb{N}$ such that for all $\mathbf{t}_m$, $1/L \leq \|\nabla f(\mathbf{t}_m)\|_{\mathrm{tr}} \leq 2$, and assume further there is some D such that $\forall k \leq K, n \in \mathbb{N} : d_{\mathrm{tr}}(\mathbf{t}_n, \mathbf{x}_k) \leq D$. We define $a_m = \alpha m^{-1/2}\|\nabla f(\mathbf{t}_m)\|_{\mathrm{tr}}$. Then for all m satisfying*

$$1 < \xi(m) := \frac{\sqrt{m+1}}{2} - \frac{KDNL(N-1)}{8\alpha} + \sqrt{\left(\frac{\sqrt{m+1}}{2} - \frac{KDNL(N-1)}{8\alpha}\right)^2 + LKN - 1},$$

we have that:

$$d_{\mathrm{tr}}(\mathbf{t}_m, V) \leq \frac{2\alpha N}{\sqrt{m+1} - KDNL(N-1)/4\alpha - 1} = \epsilon_m.$$

In particular, if f is a $\triangle_{\min}$-star-quasi-convex function, then there is some global minimum $\mathbf{t}^*$ such that

$$d_{\mathrm{tr}}(\mathbf{t}_m, \mathbf{t}^*) \leq \frac{2\alpha N}{\sqrt{m+1} - DNL(N-1)/4\alpha - 1}.$$

Proof We fix m and define:

$$M_1 = \left\lfloor \xi(m)^2 \right\rfloor.$$

We note that:

$$0 \leq \sqrt{M_1} \leq \xi(m)$$

So $\sqrt{M_1}$ lies between the roots of the polynomial

$$p(t) = \frac{N}{L}t^2 + \left(\frac{KDN^2(N-1)}{4\alpha} - \frac{N\sqrt{m+1}}{L}\right)t + \frac{N - LKN^2}{L}.$$

Therefore:

$$\begin{aligned}
\frac{N}{L}(M_1+1) + \left(\frac{KDN^2(N-1)}{4\alpha} - \frac{N\sqrt{m+1}}{L}\right)\sqrt{M_1} - KN^2 \leq 0,\\
\frac{N\sqrt{M_1}\sqrt{M_1+1}}{L} + \frac{KDN^2(N-1)\sqrt{M_1}}{4\alpha} - \frac{N\sqrt{M_1}\sqrt{m+1}}{L} \leq KN^2,\\
\frac{KDN(N-1)}{2} + 2\alpha L^{-1}(\sqrt{M_1+1} - \sqrt{m+1}) \leq \frac{2\alpha KN}{\sqrt{M_1}}.
\end{aligned}$$

Claim 38. *For all $n \geq M_1$ and m such that $1 < \xi(m)$, we have $a_n \leq \epsilon_m/N$.*

Proof We have assumed that $1 < \xi(m)$, so:

$$\begin{aligned}
M_1 &\geq \xi(m)^2 - 1,\\
&\geq (\xi(m) - 1)^2,\\
\sqrt{M_1} &\geq \xi(m) - 1, \geq (\sqrt{m+1} - KDNL(N-1)/4\alpha) - 1.
\end{aligned}$$

Hence, by the step size definition:

$$\begin{aligned}
a_n &\leq \frac{2\alpha}{\sqrt{M_1}},\\
&\leq \frac{2\alpha}{\sqrt{m+1} - KDNL(N-1)/4\alpha - 1} = \epsilon_m/N.
\end{aligned}$$

The claim is proved. □

Claim 39. *For all $n \geq m$, we have* $\max_j \Delta_j(\mathbf{t}_{M_1}) + s_{M_1} - s_m \leq K\epsilon_m$.

Proof We note that

$$\begin{aligned}
\max_j \Delta_j(\mathbf{t}_{M_1}) &\leq \sum_j \Delta_j(\mathbf{t}_{M_1}),\\
&= \sum_{k\leq K} \sum_{i \leq j \leq N} |t_{M_1 j} - x_{kj} - t_{M_1 i} + x_{ki}|,\\
&\leq \sum_{k\leq K} \sum_{i \leq j \leq N} d_{\mathrm{tr}}(\mathbf{t}_{M_1}, \mathbf{x}_k),\\
&\leq \frac{KDN(N-1)}{2}.
\end{aligned}$$

The partial sums are bounded by:

$$
\begin{aligned}
s_{M_1} - s_m &\leq \sum_{n=M_1+1}^{m} -a_n \\
&\leq \sum_{n=M_1+1}^{m} \frac{-\alpha}{Ln^{1/2}} \\
&\leq \sum_{n=M_1+1}^{m} -2\alpha L^{-1}(\sqrt{n+1} - \sqrt{n}) \\
&\leq -2\alpha L^{-1}(\sqrt{m+1} - \sqrt{M_1+1})
\end{aligned}
$$

Therefore:

$$
\begin{aligned}
\max_i \Delta_i(\mathbf{t}_{M_1}) + s_{M_1} - s_m &\leq KDN(N-1)/2 + 2\alpha L^{-1}(\sqrt{M_1+1} - \sqrt{m+1}), \\
&\leq \frac{2\alpha KN}{\sqrt{M_1}}, \\
&\leq K\epsilon_m.
\end{aligned}
$$

□

Hence by Theorem 29, we have that

$$d_{\mathrm{tr}}(\mathbf{t}_m, V) \leq \epsilon_m.$$

The bound for the tropically quasi-convex case follows from Corollary 34. □

4 Numerical Experiments

In this section we compare the practical performance of classical descent, tropical descent, SGD, tropical SGD, Adam, Adamax, and TrAdamax, with a focus on their stability behavior. The implementation for all these experiments is available in the GitHub repository at `https://github.com/Roroast/TropicalGradDescent`.

4.1 Methods

We begin by outlining our implementation of steepest descent, stochastic gradient descent (SGD), and Adam as well as their tropical variants. We take a heuristic approach to the tropicalization of SGD and Adam, replacing derivatives with tropical steepest descent directions in each case.

Steepest Descent. We will perform classical descent (CD) and tropical descent (TD) using equivalent step size sequences a_n for both methods, which we fix as

$$a_n = \frac{\gamma\|\nabla f(\mathbf{t}_n)\|}{\sqrt{n}},$$

where $\|\cdot\|$ is the Euclidean norm for classical descent and the tropical norm for tropical descent. When $\nabla f(\mathbf{t})$ is bounded this is a well-behaved step rule; a_n will converge to 0. Furthermore, for the piecewise linear problems, $\|\nabla f(\mathbf{t}_n)\|$ is bounded away from 0 almost everywhere which is not a global minimum. The convergence results of Subsection 3.2 will therefore apply.

Stochastic Gradient Descent. Stochastic gradient descent is a variant of gradient descent which looks to improve computational speed by approximating $\nabla f(t)$; rather than computing the loss and its derivative at $\mathbf{t}$ with respect to the entire dataset $f(\mathbf{t}; X)$, we evaluate the loss and its derivative with respect to a random sample $\mathbf{x}_n \in X$, $f(\mathbf{t}; \mathbf{x}_n)$. This is simplified with the notation $f_n(\mathbf{t})$. As with gradient descent, we use the step size sequence give by

$$a_n = \frac{\gamma\|\nabla f_n(\mathbf{t}_n)\|}{\sqrt{n}},$$

where $\|\cdot\|$ is the Euclidean norm for stochastic gradient descent (SGD) and the tropical norm for tropical stochastic gradient descent (TSGD).

Adam Variants. The Adam algorithm (Kingma & Ba, 2015) uses gradient momentum to compound consistent but small effects. This is done by taking first and second moments of the gradient over all steps using exponentially decaying weights. The Adamax algorithm is a variant of the Adam algorithm motivated by the use of the ∞-moments rather than second moments.

Algorithm 2 TrAdamax

$\mathbf{m}, \mathbf{u} \leftarrow \mathbf{0}$;
for $n <$ max_steps **do**
  $\mathbf{g}_{ni} \leftarrow \nabla f_i(\mathbf{t}_{n-1})$;
  $\mathbf{d}_{ni} \leftarrow (\max_j(\mathbf{g}_{nj}) - \min_j(\mathbf{g}_{nj})) * \mathbf{1}_{\mathbf{g}_{ni}>0}$;
  $\mathbf{m}_n \leftarrow \beta_1 * \mathbf{m}_{n-1} + (1-\beta_1) * \mathbf{d}_n$;
  $\mathbf{u}_n \leftarrow \max(\beta_2 * \mathbf{u}_{n-1}, |\mathbf{d}_n|)$;
  $\mathbf{t}_n \leftarrow \mathbf{t}_n - \alpha/(1-\beta_1^n) * \mathbf{m}_n/\mathbf{u}_n$;

To adapt the Adam algorithm for the tropical setting, we take moments of an unnormalized tropical descent direction. When looking to take higher moments of the gradient over n, the L_∞ is the natural option for the tropical setting. We therefore look to use the Adamax recursion relation on the ∞-moment estimate $\mathbf{u}_n$:

$$\mathbf{u}_{n+1} = \max(\beta_2 \mathbf{u}_n, \mathbf{d}_n).$$

The tropicalized Adamax algorithm is then given by Algorithm 2.

4.2 Computation

We now look to solve our centrality statistics, Wasserstein projection problems, and tropical linear regression using the 7 gradient methods discussed: classical descent (CD), tropical descent (TD), stochastic gradient descent (SGD), tropical stochastic gradient descent(TSGD), Adam, Adamax, and TrAdamax. We run computations for 12 datasets of different sizes, shapes and dimensions; motivated by phylogenetic data science applications, we consider datasets generated by branching processes, a coalescent processes, or Gaussian distributions on $\mathbb{R}^N/\mathbb{R}\mathbf{1}$, for $N = 6, 28$ and samples of size 10 or 100. We take same random initializations for each gradient method, taken from a Gaussian distribution on $\mathbb{R}^N$. Appendix B provides further details on the simulated datasets and computational methodology used. For each gradient method, we use learning rates which have been optimized for each dataset size and dimension, across different distribution types (Appendix C).

Rather than investigating the relative convergence rates of each method, we are interested in the relative stability of local minima. We therefore plot the CDF of the relative log error after 1000 steps over 50 random initializations, which can highlight certain error values at which several initializations have accumulated. When the CDF of one method dominates another, it has a higher probability of achieving the same error and is therefore a more suitable method.

As we will observe, tropical gradient methods are stable at fewer points than classical gradient methods; as a result, tropical methods typically have further to travel before reaching a stable point. Over few steps, this would be compensated for by selecting larger learning rates for tropical gradient methods relative to classical gradient methods, leading to less precise results in the case of tropical gradient methods. Our computations are run over 1000 steps, as this gives enough time for tropical gradient methods to stabilise, given our initialisations, without sacrificing learning rate; we see in Appendix C that neither classical nor tropical learning rates are consistently greater than the other.

4.2.1 Centrality Statistics

We first consider the behavior of tropical descent methods for centrality statistics, Fermat–Weber points and Fréchet means. These optimization problems are classically convex problems, as well as being both max-tropical and min-tropical location problems, and therefore lend themselves well to gradient methods.

The experimental behavior of these problems is similar, so in this section we include the results for the Fermat–Weber problem while the results for Fréchet means can be found in Appendix E.

Figure 5 shows the CDF of the log relative error over 1000 steps of each gradient method for the 12 datasets. We note that the deterministic methods achieve similar results for most datasets, while the stochastic methods achieve the worst error for all 12 datasets and TSGD is performing worse than SGD. The disparity between SGD and TSGD is greatest when the dimensionality of the data is greater than the size of the dataset, and for coalescent data. We see similar behavior in the log error of Tropical Descent and TrAdamax; they perform on par with their classical counterparts for each dataset other than the 28 by 10 coalescent data, for which the tropical gradient methods are not achieving the same accuracy. While it is unclear why this is, we note that coalescent data is supported on the space of ultrametric trees, which is a tropical hyperplane (Ardila & Klivans, 2006; Page *et al.*, 2020) and may influence the uniqueness of minima.

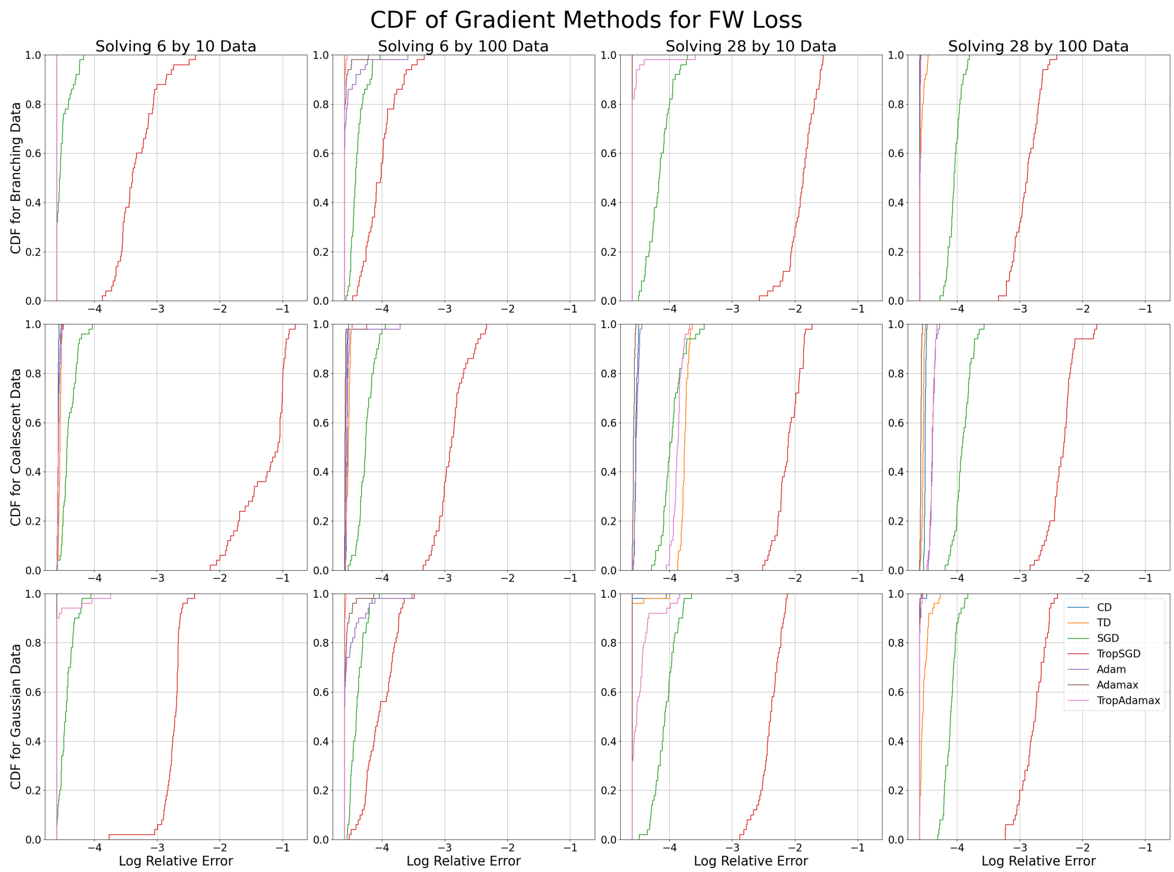

Figure 5: The CDF of the log relative error after 1000 steps for each gradient method across 50 initializations when minimizing the Fermat–Weber objective. Each sub-figure corresponds to a dataset of different dimensionality (columns) and distribution (rows).

4.2.2 Wasserstein Projections

The Wasserstein projection problems highlight the differences in convergent behavior between classical and tropical gradient methods, as they are min-tropical location problems but not classically convex. The CDFs of the final log relative errors can be found in Figures 6 and 7.

In Figures 6 and 7 we see large steps at various intervals in the CDF plots, which is indicative of stable local minima. This is particularly prevalent in low dimensional space. Classical descent, Adam, and Adamax are particularly susceptible to this, while tropical descent and TrAdamax are more likely to pass these local minima. This difference is most pronounced when solving the ∞-Wasserstein problem in low dimensions—we

Table 1: The mean log relative error of each gradient method after 1000 steps for each dataset when minimizing the Fermat–Weber objective function. The minimal mean errors for each dataset are in bold.

Data	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6×10 Branching Data	**-4.60**	**-4.60**	-4.50	-3.32	**-4.60**	**-4.60**	**-4.60**
6×10 Coalescent Data	**-4.58**	-4.55	-4.40	-1.28	-4.57	-4.57	-4.57
6×10 Gaussian Data	**-4.60**	**-4.60**	-4.44	-2.75	**-4.60**	**-4.60**	-4.56
6×100 Branching Data	**-4.60**	**-4.60**	-4.39	-4.02	-4.54	-4.58	-4.59
6×100 Coalescent Data	-4.55	-4.54	-4.25	-2.88	-4.56	**-4.58**	-4.55
6×100 Gaussian Data	**-4.60**	**-4.60**	-4.39	-4.03	-4.52	-4.57	-4.59
28×10 Branching Data	**-4.60**	**-4.60**	-4.15	-1.89	**-4.60**	**-4.60**	-4.56
28×10 Coalescent Data	-4.53	-3.76	-3.96	-2.11	-4.52	**-4.57**	-3.87
28×10 Gaussian Data	-4.58	-4.58	-4.07	-2.40	**-4.60**	**-4.60**	-4.47
28×100 Branching Data	**-4.59**	-4.57	-4.04	-2.89	**-4.59**	**-4.59**	**-4.59**
28×100 Coalescent Data	-4.51	-4.54	-3.91	-2.33	-4.39	**-4.58**	-4.39
28×100 Gaussian Data	**-4.59**	-4.53	-4.10	-2.78	**-4.59**	**-4.59**	**-4.59**

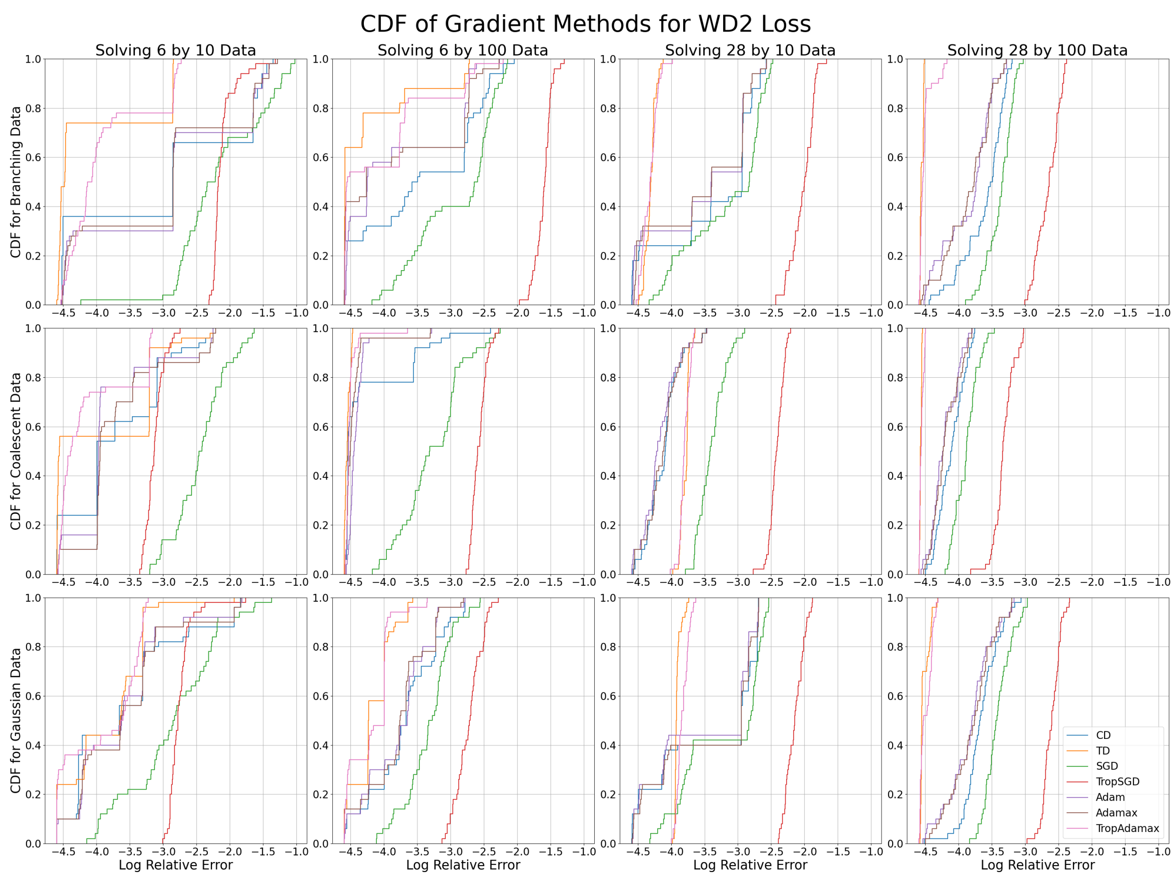

Figure 6: The CDF of the log relative error after 1000 steps for each gradient method across 50 initializations when minimizing the 2-Wasserstein projection objective. Each sub-figure corresponds to a dataset of different dimensionality (columns) and distribution (rows).

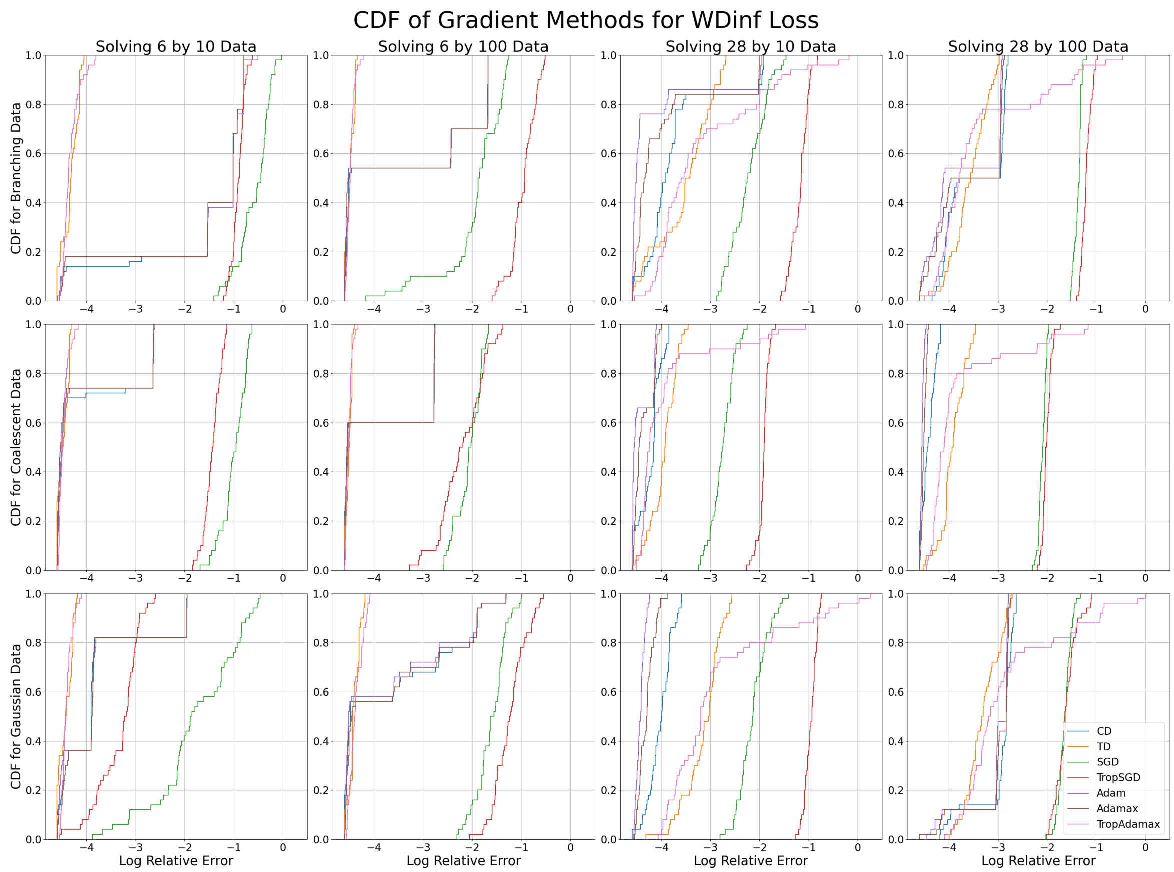

Figure 7: The CDF of the log relative error after 1000 steps for each gradient method across 50 initializations when minimizing the ∞-Wasserstein projection objective. Each sub-figure corresponds to a dataset of different dimensionality (columns) and distribution (rows).

are almost certain to reach the minimum with small error, while classical methods have a 50-80% chance of achieving the same error.

4.2.3 Linear Regression

As the coalescent data lies on a tropical hyperplane (Ardila & Klivans, 2006; Page *et al.*, 2020), the minimum of the linear regression function for such data is exactly 0. Figure 8 therefore shows the CDF of the log error rather than the CDF of the log relative error.

Table 2 shows the mean log errors of each gradient method and dataset, from which we see that the tropical methods outperform their classical counterparts for all but one dataset, often by a factor of two. In Figure 8 we see that the CDF of log errors for tropical descent and TrAdamax consistently dominates those of classical descent, Adam and Adamax, to the extent that the support of their CDFs does not intersect that of their classical counterparts for some datasets. We also note that unlike the previous problems, TSGD is outperforming SGD when solving the linear regression problem, often to the extent that TSGD outperforms all classical gradient methods.

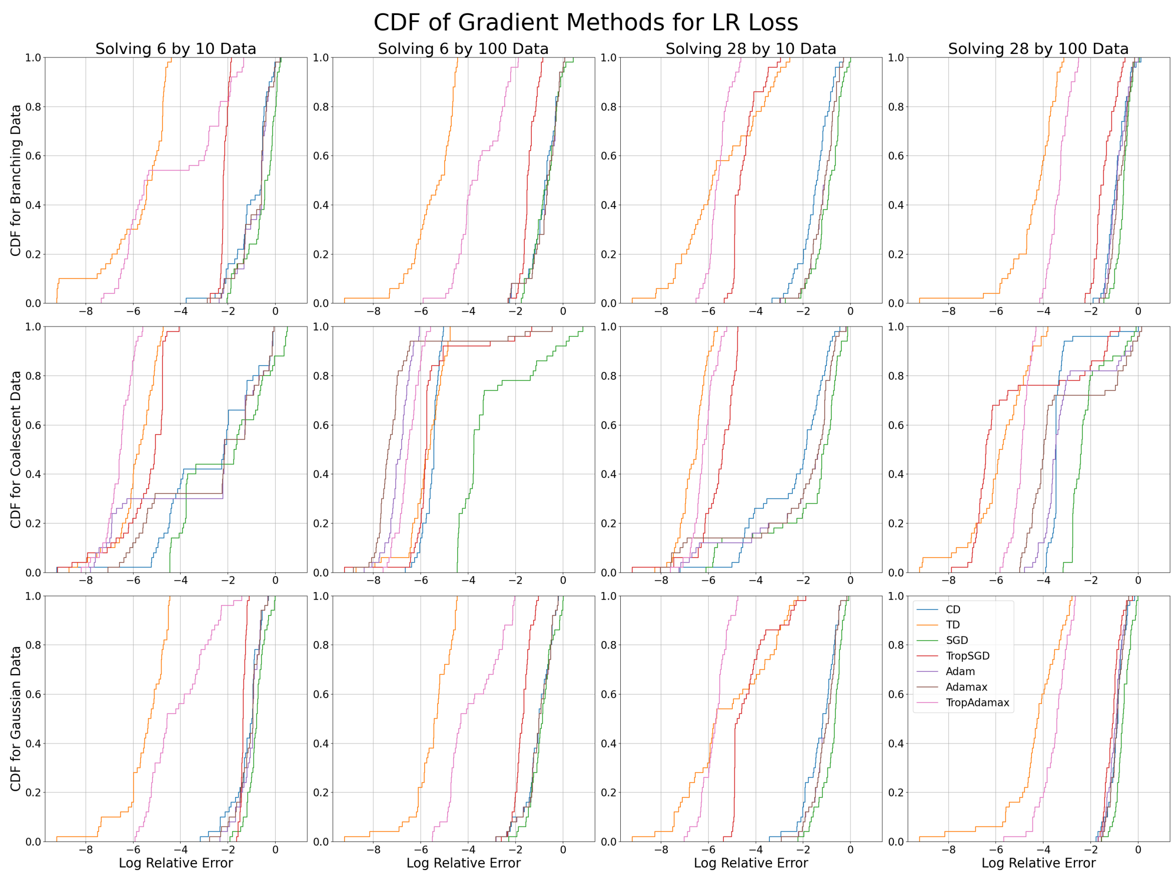

Figure 8: The CDF of the log relative error after 1000 steps for each gradient method across 50 initializations when minimizing the linear regression objective. Each sub-figure corresponds to a dataset of different dimensionality (columns) and distribution (rows).

5 Conclusion and Discussion

We have introduced the concept of tropical descent as the steepest descent method with respect to the tropical norm. We have demonstrated, both theoretically and experimentally, that it possesses well-behaved

Table 2: The mean log relative error of each gradient method after 1000 steps for each dataset when minimizing the linear regression objective function. The minimal mean errors for each dataset are in bold.

Data	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6×10 Branching Data	-0.97	**-5.79**	-0.56	-2.16	-0.80	-0.84	-4.39
6×10 Coalescent Data	-2.70	-5.87	-2.11	-5.47	-2.97	-2.66	**-6.58**
6×10 Gaussian Data	-1.19	**-5.48**	-0.80	-1.35	-1.05	-1.10	-4.10
6×100 Branching Data	-0.79	**-5.49**	-0.71	-1.43	-0.72	-0.72	-3.53
6×100 Coalescent Data	-5.54	-5.76	-3.11	-5.57	-6.90	**-7.01**	-6.52
6×100 Gaussian Data	-1.04	**-5.50**	-0.88	-1.70	-1.02	-1.02	-3.81
28×10 Branching Data	-1.49	-5.51	-0.93	-4.49	-1.14	-1.14	**-5.57**
28×10 Coalescent Data	-2.43	**-6.63**	-1.77	-5.55	-2.16	-2.29	-6.18
28×10 Gaussian Data	-1.26	-5.16	-0.79	-4.29	-1.08	-1.08	**-5.72**
28×100 Branching Data	-0.90	**-4.37**	-0.61	-1.42	-0.85	-0.77	-3.32
28×100 Coalescent Data	-3.33	**-5.84**	-2.17	-5.35	-3.02	-3.20	-4.93
28×100 Gaussian Data	-0.91	**-4.35**	-0.62	-1.07	-0.96	-0.90	-3.51

properties with respect to a wide class of optimization problems on the tropical projective torus, including several key statistical problems for the analysis of phylogenetic data. As well as proposing our own framework for tropical gradient methods, this work also provides a detailed review of the varying behavior of popular gradient methods when applied to tropical statistical optimization problems.

Our theoretical results outline the relative strengths of tropical descent over classical descent for tropical location problems, but only provide global solvability for 1-sample problems. In practice, our experimental results go further; the optimization problems studied are location problems, rather than $\triangle$-star-quasi-convex problems, and yet tropical descent appears to consistently converge to a global minimum, not just the tropical convex hull of our dataset. It would therefore be a natural extension of this work to identify how much we can relax the $\triangle$-star-quasi-convexity condition while guaranteeing convergence.

In this paper we have considered tree data with 4 to 8 leaves, and samples of size 10 to 100—at this scale, we have observed (T/)SGD to be a significant sacrifice in accuracy with little benefit in computational time. Furthermore, at this scale, the relative merits of SGD versus its tropical counterpart are inconsistent; TSGD out-performs SGD when solving the linear regression problem, but performs worse when finding centrality statistics. In a similar vein, initial investigations suggest that the classical stochastic algorithm for the computation of Fréchet means—Sturm's algorithm (Sturm, 2003)—performs poorly for the computation of tropical Fréchet means due to the inconsistent curvature of the tropical projective torus (Amndola & Monod, 2023). However, for applications to larger datasets and more complex objective functions, stochastic gradient methods may become necessary. It would then be important to understand the theoretical differences between SGD and TSGD in terms of their convergence guarantees in the tropical setting and their comparative performance at different scales of data dimension.

The recent identification of tropical rational functions and ReLU neural networks (Zhang *et al.*, 2018) has inspired new avenues of research in tropical applications, into the tropical geometry of neural networks (Brandenburg *et al.*, 2024; Yoshida *et al.*, 2024). It is therefore natural to consider the possible extensions of this work to neural network training; can we identify local tropical quasi-convexity in the loss functions of ReLU neural networks, and how can we extend tropical steepest descent to the space of tropical rational functions?

Finally, as shown by (Akian *et al.*, 2023), the tropical linear regression problem is polynomial-time equivalent to mean payoff games—two-player perfect information games on a directed graph (Ehrenfeucht & Mycielski, 1979; Gurvich *et al.*, 1988). This family of problems is of particular interest in game theory as they are known to be NP $\cap$ co-NP. In our numerical experiments, tropical gradient methods demonstrate a particularly strong performance in solving tropical linear regression, and while our methods are designed to approximate the minimum rather than compute it exactly, tropical descent may prove to be an efficient tool when solving large-scale mean payoff games.

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Declarations

The authors have no relevant financial or non-financial interests to disclose. The authors have no competing interests to declare that are relevant to the content of this article. All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. The authors have no financial or proprietary interests in any material discussed in this article.

A PCA and Logistic Regression

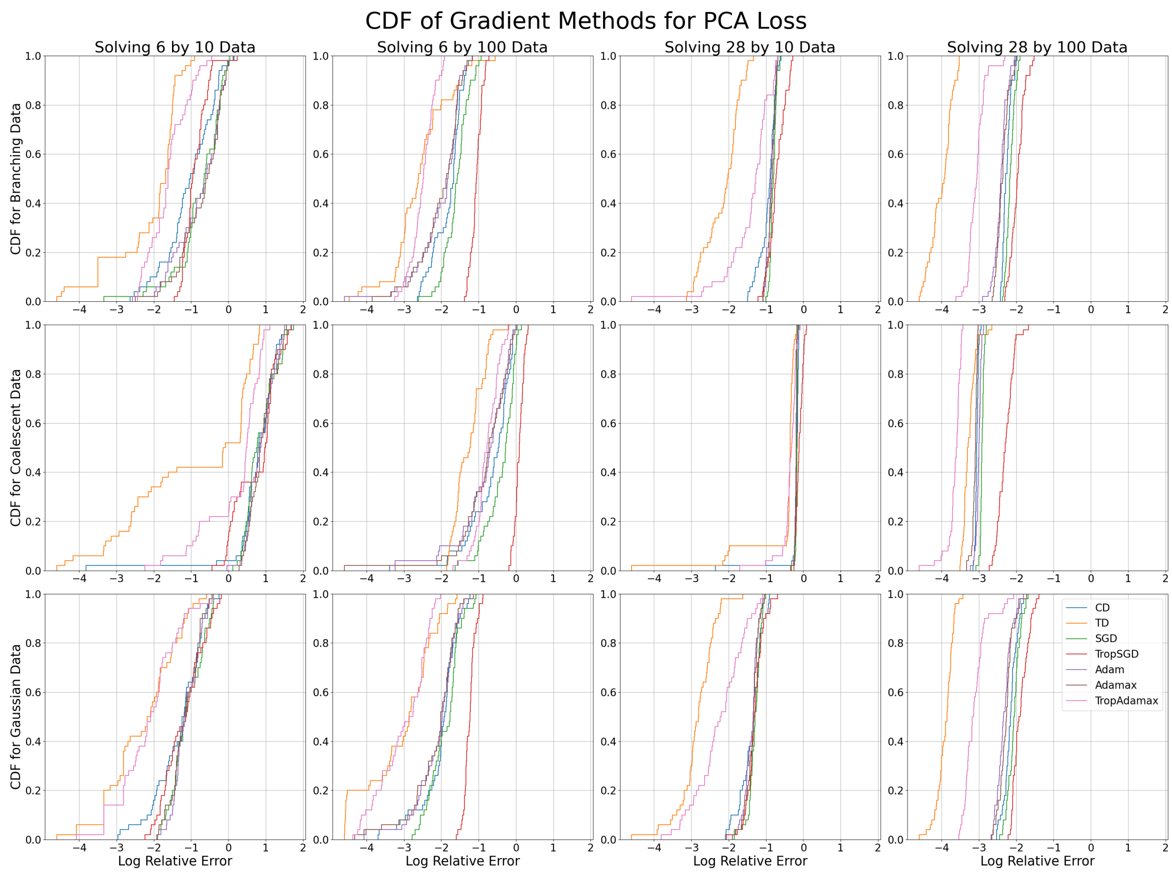

Figure 9: The CDF of the log relative error after 1000 steps for each gradient method across 50 initializations when minimizing the PCA objective. Each sub-figure corresponds to a dataset of different dimensionality (columns) and distribution (rows).

As well as the location problems outlined in this paper, we performed the same numerical experiments with the tropical polytope PCA loss function (Yoshida *et al.*, 2019) and tropical logistic regression loss function (Aliatimis *et al.*, 2024). These loss functions take multiple points in $\mathbb{R}^N/\mathbb{R}\mathbf{1}$ as argument, and we execute our gradient methods on each argument simultaneously. Figures 9 and 10 show the resulting CDF of

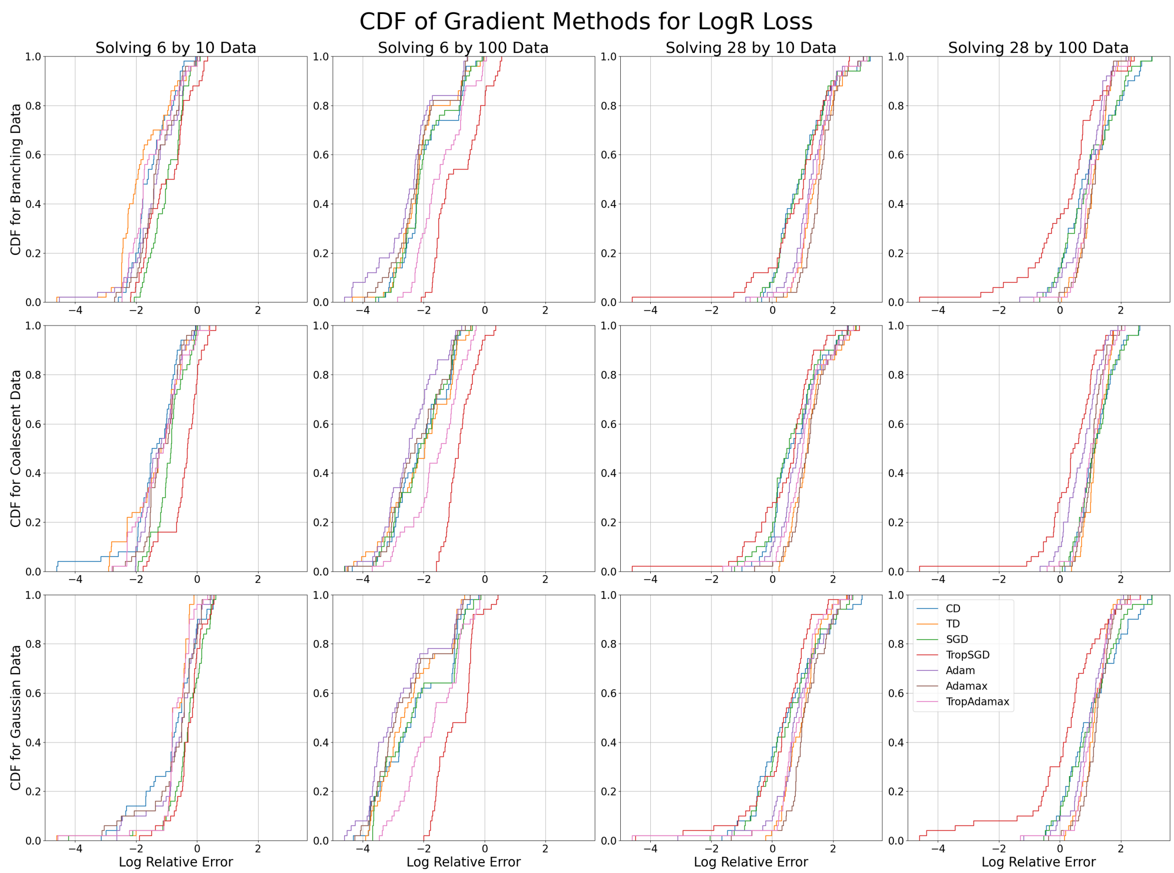

Figure 10: The CDF of the log relative error after 1000 steps for each gradient method across 50 initializations when minimizing the logistic regression objective. Each sub-figure corresponds to a dataset of different dimensionality (columns) and distribution (rows).

our gradient methods for these problems. These problems do not lend themselves well to gradient methods, as the relative log error is rather large for all of our proposed methods. However, for both problems the tropical descent and TrAdamax methods are outperforming the classical methods quite consistently.

B Methodology

The same 12 datasets are used for each experiment in Section 4. These datasets live in $\mathbb{R}^6/\mathbb{R}\mathbf{1}$ or $\mathbb{R}^{28}/\mathbb{R}\mathbf{1}$; these dimensions correspond to the spaces of trees with 4 and 8 leaves respectively. Branching process datasets are generated using the `R` package `ape` (Paradis & Schliep, 2019) with exponentially distributed branch lengths—this produces data supported on the space of metric trees, i.e., the tropical Grassmannian. Coalescent data has also been generated using `ape` and is supported on the space of ultrametric trees, which is a tropical linear space. Gaussian data is generated using `NumPy` (Harris *et al.*, 2020), and is supported on the general tropical projective torus.

In the computation of the Wasserstein projection objectives, we require a secondary dataset of different dimensionality; for each of the 12 datasets, we generate datasets of the same size and distribution, but in $\mathbb{R}^3/\mathbb{R}\mathbf{1}$ or $\mathbb{R}^{21}/\mathbb{R}\mathbf{1}$ which contain the spaces of trees with 3 and 7 leaves respectively.

We use `PyTorch` (Paszke *et al.*, 2019) for the computation and differentiation of each objective function. Max-tropical descent directions are used for tropical gradient methods; we therefore optimize the min-tropical location problems—Wasserstein projections and tropical linear regression—with respect to $-\mathbf{t}$.

C Learning Rate Tuning

The Adam algorithms — Adam, Adamax and TrAdamax — are run using the default $\beta = (0.9, 0.999)$ parameters. We then tune the learning rates used for each method; Gradient Descent, Tropical Descent, SGD, Tropical SGD, Adam, Adamax, and TrAdamax. We perform an initial test of learning rates γ on a logarithmic scale from 0.0001 to 30 over 1000 steps for each algorithm, with 5 random initializations for each dataset. From this, we narrowed our search to take 18 initializations at 0.5 intervals on the log scale in near optimal regions for each problem, gradient method, and data dimensionality. We then selected the learning rate with minimal mean log relative error in each case; Tables 3 to 9 show the selected learning rates.

Table 3: Learning rates for each gradient method when solving the Fermat–Weber problem.

Data Dim	Data Count	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6	10	0.135	0.135	0.135	0.135	0.00674	0.00674	0.00674
	100	0.135	0.135	0.135	0.135	0.00248	0.00248	0.00674
28	10	0.368	0.368	0.368	1	0.00674	0.00674	0.0183
	100	0.368	0.135	0.368	1	0.00674	0.00674	0.0183

Table 4: Learning rates for each gradient method when solving the Fréchet mean problem.

Data Dim	Data Count	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6	10	0.135	0.135	0.135	0.135	0.00674	0.00674	0.00674
	100	0.135	0.135	0.135	0.135	0.00674	0.00248	0.00674
28	10	0.368	0.368	0.368	1	0.00674	0.00674	0.0183
	100	0.368	0.135	0.368	1	0.00674	0.00674	0.0183

Table 5: Learning rates for each gradient method when solving the 2-Wasserstein projection problem.

Data Dim	Data Count	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6	10	2.72	1	1	0.135	0.135	0.368	0.368
	100	2.72	2.72	0.368	0.135	0.135	0.135	0.135
28	10	0.368	0.368	0.368	1	0.00674	0.0183	0.0183
	100	2.72	2.72	0.368	1	0.0498	0.135	0.0498

Table 6: Learning rates for each gradient method when solving the ∞-Wasserstein projection problem.

Data Dim	Data Count	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6	10	0.0498	0.135	0.0498	0.368	0.00674	0.00674	0.0183
	100	0.0498	0.135	0.135	2.72	0.00674	0.00674	0.0183
28	10	0.368	1	0.368	1	0.00674	0.0183	0.0498
	100	0.368	1	0.368	1	0.00674	0.0183	0.0498

Table 7: Learning rates for each gradient method when solving the polytope PCA problem.

Data Dim	Data Count	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6	10	2.72	2.72	1	1	0.135	0.135	0.368
	100	2.72	7.39	1	1	0.135	0.135	1
28	10	7.39	7.39	0.368	1	0.0183	0.0183	0.135
	100	2.72	7.39	0.368	1	0.0498	0.0498	0.0498

Table 8: Learning rates for each gradient method when solving the logistic regression problem.

Data Dim	Data Count	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6	10	7.39	2.72	1	2.72	0.368	1	0.0498
	100	1	0.368	1	2.72	0.00248	0.00248	0.00248
28	10	1	0.368	1	2.72	0.00248	0.00248	0.00674
	100	1	0.368	1	7.39	0.00248	0.00248	0.00674

Table 9: Learning rates for each gradient method when solving the linear regression problem.

Data Dim	Data Count	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6	10	0.368	0.135	0.368	0.135	0.00248	0.0183	0.00674
	100	0.135	0.135	0.368	0.0498	0.00248	0.00248	0.00674
28	10	0.368	0.0498	0.0498	0.135	0.00248	0.00248	0.0183
	100	1	0.368	1	7.39	0.0498	0.0498	0.0498

D Runtimes

Experiments were implemented on a AMD EPYC 7742 node using a single core, 8 GB. Table 10 shows the average time taken per initialization for each gradient method and optimization problem.

Table 10: Average runtimes for 1000 steps of each gradient method when minimizing each objective function.

Stats Prob	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
Fermat–Weber	0.480	0.520	0.435	0.499	0.391	0.460	0.464
Fréchet Mean	0.347	0.363	0.301	0.342	0.294	0.422	0.370
2-Wasserstein Projection	2.739	2.134	2.392	2.209	2.341	2.246	2.272
∞-Wasserstein Projection	2.947	3.049	2.772	2.565	2.501	3.162	2.807
Polytope PCA	1.117	1.081	1.014	1.052	1.037	1.174	1.029
Logistic Regression	0.987	1.015	0.962	0.934	0.943	1.084	0.943
Linear Regression	0.494	0.499	0.386	0.420	0.386	0.537	0.433

E Further Figures

E.1 Fréchet Means

Here we include the results for experiments run for the computation of Fréchet means.

Table 11: The mean log relative error of each gradient method after 1000 steps for each dataset when minimizing the Fréchet mean objective function. The minimal mean errors for each dataset are in bold.

Data	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6×10 Branching Data	**-4.59**	-4.54	-3.98	-2.99	-4.58	**-4.59**	-4.53
6×10 Coalescent Data	**-4.58**	-4.47	-4.07	-1.82	-4.57	**-4.58**	-4.45
6×10 Gaussian Data	**-4.59**	-4.33	-4.09	-2.63	-4.58	**-4.59**	-4.35
6×100 Branching Data	**-4.59**	**-4.59**	-4.35	-4.03	**-4.59**	-4.58	**-4.59**
6×100 Coalescent Data	**-4.57**	-4.56	-4.36	-3.29	-4.55	-4.56	**-4.57**
6×100 Gaussian Data	-4.59	**-4.60**	-4.39	-4.12	-4.59	-4.58	-4.59
28×10 Branching Data	**-4.58**	-3.73	-3.78	-1.98	-4.55	**-4.58**	-3.83
28×10 Coalescent Data	-4.56	-3.67	-3.98	-2.46	-4.53	**-4.57**	-3.70
28×10 Gaussian Data	-4.55	-4.07	-3.91	-2.41	-4.55	**-4.57**	-4.04
28×100 Branching Data	**-4.59**	-4.58	-4.03	-2.98	-4.58	**-4.59**	-4.57
28×100 Coalescent Data	-4.54	-4.56	-4.08	-2.66	-4.47	**-4.57**	-4.46
28×100 Gaussian Data	**-4.59**	-4.56	-4.08	-2.78	**-4.59**	**-4.59**	-4.58

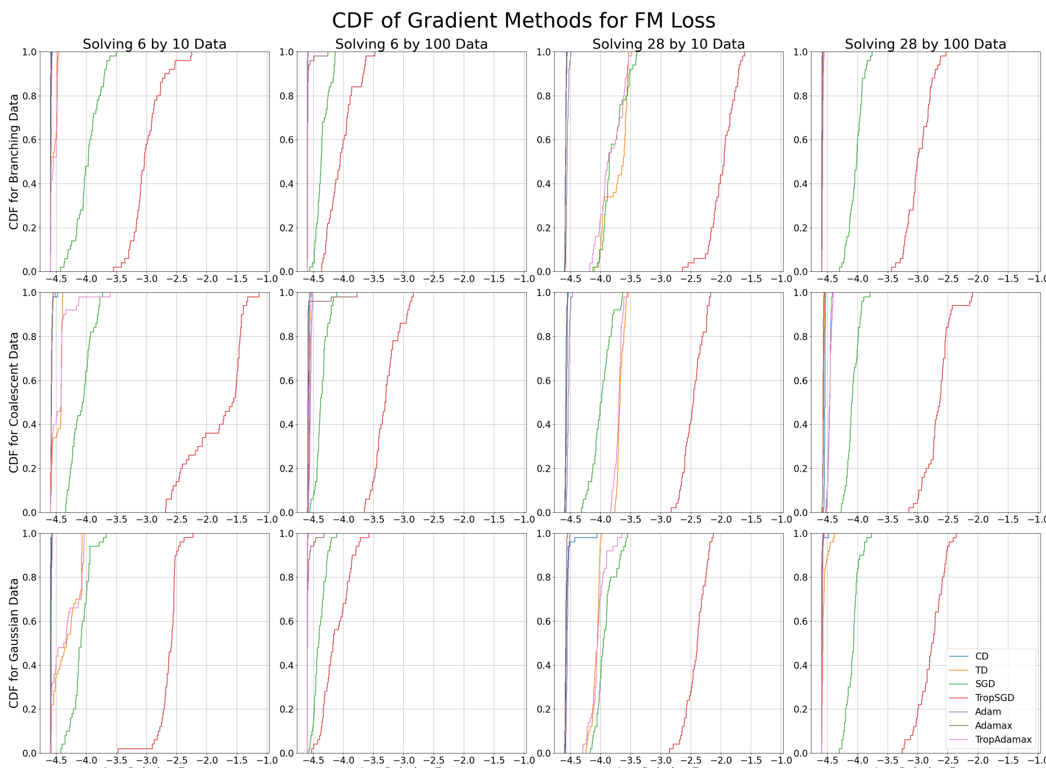

Figure 11: The CDF of the log relative error after 1000 steps for each gradient method across 50 initializations when minimizing the Fréchet mean objective. Each sub-figure corresponds to a dataset of different dimensionality (columns) and distribution (rows).

E.2 Wasserstein Projections

From Tables 12 and 13, we see that when solving the 2-Wasserstein projection problem, tropical methods are optimal in all cases other than 28×10 coalescent data. For the ∞-Wasserstein projection problem in small dimensions, tropical descent outperforms other methods by some margin, while the comparative performances are less consistent for the ∞-Wasserstein projection problem in high dimensions.

Table 12: The mean log relative error of each gradient method after 1000 steps for each dataset when minimizing the 2-Wasserstein projection objective function. The minimal mean errors for each dataset are in bold.

Data	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6×10 Branching Data	-3.02	**-4.09**	-2.16	-2.11	-2.95	-3.00	-3.89
6×10 Coalescent Data	-3.71	-3.92	-2.44	-3.13	-3.78	-3.68	**-4.12**
6×10 Gaussian Data	-3.59	-3.83	-2.87	-2.75	-3.61	-3.57	**-3.89**
6×100 Branching Data	-3.44	**-4.24**	-2.91	-1.60	-3.79	-3.79	-4.02
6×100 Coalescent Data	-4.26	**-4.55**	-3.27	-2.58	-4.39	-4.45	-4.50
6×100 Gaussian Data	-3.69	-4.17	-3.33	-2.73	-3.78	-3.78	**-4.18**
28×10 Branching Data	-3.39	-4.31	-3.16	-2.03	-3.53	-3.55	**-4.32**
28×10 Coalescent Data	-4.11	-3.79	-3.39	-2.42	**-4.17**	-4.12	-3.80
28×10 Gaussian Data	-3.43	**-3.91**	-3.21	-2.15	-3.50	-3.44	-3.83
28×100 Branching Data	-3.61	**-4.55**	-3.37	-2.65	-3.85	-3.85	-4.51
28×100 Coalescent Data	-4.11	**-4.57**	-3.89	-3.32	-4.21	-4.19	-4.56
28×100 Gaussian Data	-3.66	**-4.52**	-3.41	-2.61	-3.82	-3.80	-4.49

Table 13: The mean log relative error of each gradient method after 1000 steps for each dataset when minimizing the ∞-Wasserstein projection objective function. The minimal mean errors for each dataset are in bold.

Data	CD	TD	SGD	TSGD	Adam	Adamax	TrAdamax
6×10 Branching Data	-1.63	**-4.32**	-0.57	-0.91	-1.68	-1.69	**-4.32**
6×10 Coalescent Data	-4.00	**-4.47**	-0.99	-1.43	-4.03	-4.04	-4.46
6×10 Gaussian Data	-3.77	**-4.42**	-1.76	-3.33	-3.76	-3.75	-4.40
6×100 Branching Data	-3.35	**-4.48**	-1.94	-0.95	-3.34	-3.36	-4.47
6×100 Coalescent Data	-3.85	**-4.51**	-2.06	-2.21	-3.84	-3.85	**-4.51**
6×100 Gaussian Data	-3.63	**-4.40**	-1.61	-1.22	-3.70	-3.63	-4.36
28×10 Branching Data	-3.68	-3.53	-2.22	-1.17	**-4.11**	-3.93	-3.17
28×10 Coalescent Data	-4.22	-3.95	-2.76	-1.92	**-4.42**	-4.36	-3.94
28×10 Gaussian Data	-4.02	-3.14	-2.13	-0.95	**-4.43**	-4.30	-2.85
28×100 Branching Data	-3.46	-3.58	-1.37	-1.19	**-3.68**	-3.59	-3.39
28×100 Coalescent Data	-4.41	-3.92	-2.09	-2.00	**-4.54**	-4.51	-3.81
28×100 Gaussian Data	-2.98	**-3.30**	-1.64	-1.61	-3.07	-3.04	-2.80

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